

ACHINTYA RAI

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EDUCATION

University of California, San Diego
Bachelor of Science in Computer Engineering

Sept. 2024 – June 2028
GPA: 3.9/4.0

WORK EXPERIENCE

Adobe

Software Engineering Intern

San Jose, CA

June 2026 – Sept. 2026

- Automated distributed data-processing systems in **Databricks**, profiling **60.6M+** payment events to surface **86,387** duplicate records and a **47%** null rate that fixed the downstream data source.
- Designed a time-series anomaly detector, iterating over numerical algorithms to select a model handling multi-layer seasonality across **10+** workflows and **30+** countries.
- Built SQL/MLOps pipelines automating weekly retraining and hourly inference end-to-end.
- Deployed monitoring services via **Jenkins** CI/CD, debugging system-level scheduling failures.

VALT Health

Founding Software Engineer

Remote

Aug. 2025 – Feb. 2026

- Engineered a Flask backend ingesting **100k+** datapoints from fitness devices, designing database schemas to unify data formats across **20+** distinct sources.
- Implemented asynchronous batch processing and storage workflows in **GCP**, yielding a **35%** increase in execution speed and **60%** reduction in redundant database writes.
- Automated environment setup and data seeding via **Bash** scripts, cutting testing time by **80%**.

RESEARCH

Graph Theory & Neural Architecture

Undergraduate Research Fellow (NSF REU)

San Diego, CA

March 2026 – Present

- Provisioned virtualized GPU environments to run **DREAMPlace** placement workloads, isolating dependencies to reproduce large-scale experiments across shared research hardware.
- Engineered a Python pipeline applying linear-algebra weight-matrix extraction to reduce **500GB** model checkpoints to **13GB** manifolds, achieving a **30x** runtime reduction on **8GB RAM** systems.
- Applied **Rent's Rule** and Louvain clustering to lift network density from **0.18%** to **13.25%**.

Data Science Teaching & Learning Lab

Undergraduate Researcher

San Diego, CA

Sept. 2024 – Present

- Managed **Docker** containers and a compute cluster to provision dataset access for lab pipelines.
- Built a **FastAPI** tool using the Gemini API to auto-label **16k+** student AI-usage datapoints.
- Reduced labeling time by **90%** through three iterations of user-centric UI/UX refinements.

PROJECTS

SquaredUpChess | *React, TypeScript, Django, WebSockets, AWS, Docker* 

Aug. 2025 – Dec. 2025

- Containerized a full-stack scheduling platform with **Docker** on **AWS**, architecting **20+** **Django REST endpoints** for authentication, session management, and payments.
- Established a **CI/CD pipeline** with logging and automated testing for **99.9%** deploy reliability.

NanoGPT | *Python, PyTorch, Transformers, Neural Networks* 

Mar. 2026

- Implemented a transformer from scratch in **PyTorch**, sustaining **100%** GPU utilization in training.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL (PostgreSQL/SQLite), Bash, HTML/CSS

Infrastructure & Tools: Docker, Jenkins, Git, GitHub Actions, Linux, Unix/Shell Scripting, Container & Cluster Management, Virtualization, Distributed Systems

AI & Data: PyTorch, TensorFlow, NumPy, Databricks, AWS, GCP, FastAPI, Django, Node.js, Gemini/OpenAI API, MCP